

Ammar Raneez

AI ENGINEER · SOLUTIONS ARCHITECT · TECHNICAL LEAD

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SUMMARY

Gold Medal graduate, IEEE-published AI Engineer & Full-Stack Architect with 4+ years architecting cloud-based production AI systems – from real-time voice contact centres to medical-grade diagnostic platforms. Specialized in end-to-end AI pipelines, cloud-native architecture, and translating research into scalable products.

WORK EXPERIENCE

SOFTSORA **May 2025 - Present**

Head of Engineering & Tech Lead *Colombo*, Sri Lanka

- Architect and lead development of an AI-powered voice contact-centre, building a real-time conversational system using Amazon Connect, Transcribe, Streams, Python, TypeScript, Next.js, React, and Express.
- Engineered audio processing pipelines for transcription analytics, enabling automated validation and transformation of high-throughput voice data for customer support workflows with sub-2s latency.
- Designed scalable backend services and REST APIs using Express and NestJS to support high-throughput event-driven workloads and enterprise-grade integrations.
- Built semantic search and response-selection systems using Bedrock embeddings, improving response accuracy by 30%+ compared to traditional fuzzy matching approaches.
- Established full-stack architecture standards and mono-repo development patterns (NX), creating reusable boilerplates and accelerating developer onboarding by ~50%.
- Implemented automated CI/CD pipelines and Infrastructure as Code (AWS CDK) to provision cloud infrastructure and enable zero-downtime deployments across environments.
- Developed intelligent workflow automation tools including an agentic system that converts Figma designs into structured engineering user stories, reducing manual analysis effort by ~80%.
- Mentor and guide engineers across frontend and backend development, conducting architecture reviews, code reviews, and technical training on TypeScript and modern React patterns.

IIT **Jul. 2023 - Present**

Visiting Lecturer *Colombo*, Sri Lanka

- Deliver lectures to 500+ undergraduate students across Applied AI and client-server systems.
- Review & provide feedback for final-year research projects and critically evaluate them as part of the viva panel.

99X **May 2023 – Jun. 2025**

Senior Software Engineer | Jul. 2024 - Jun. 2025 *Colombo*, Sri Lanka

- Led development of a segmentation system for microscopic material analysis, designing custom CNN architectures in TensorFlow achieving 85%+ IoU accuracy on ferrite structures upon extensive tuning.
- Built a complete ML pipeline for image segmentation, including preprocessing services with FastAPI, model training with TensorFlow, experiment tracking via MLflow, and containerized deployment on Azure.

- Developed an automated computer vision labeling pipeline using OpenCV, reducing manual annotation time from 40+ hours to under 2 hours through intelligent pre-labeling and validation workflows.
- Designed a carbon credit forecasting system using LSTM neural networks, improving prediction accuracy by 50%+ compared to manual estimation methods, with automated monitoring and retraining workflows.
- Engineered scalable ML microservices using FastAPI, Docker, and Kubernetes, enabling reliable model serving and production monitoring with Prometheus and Grafana.
- Led modernization initiatives including migration of legacy applications to TypeScript and modern frontend architecture, significantly improving maintainability and reducing production bugs.

Software Engineer | May 2023 - Jul. 2024

- Developed a time-series forecasting platform using Python, FastAPI, and ARIMA (R), automating statistical analysis workflows and improving prediction accuracy by 50%+.
- Designed and implemented a data ingestion and ETL pipeline on Azure Data Factory, automating preprocessing workflows and eliminating manual data preparation.
- Prototyped computer vision spatial analysis systems using YOLO and Grounding DINO to detect objects and calculate relative surface-area metrics with 85%+ detection accuracy.
- Delivered scalable full-stack features using React, Angular, Node.js, and Express supporting micro-frontend architectures and distributed service communication.

Trainee Software Engineer | Jul. 2021 - Jul. 2022

- Played the role of the team's "bug hunter," proactively identifying and resolving full stack production bugs drastically improving system stability and performance.
- Designed and delivered a mobile application with React Native, GraphQL, and graph databases.

NIFTRON **Jan. 2021 – May 2023**

Full Stack Developer | Jul. 2022 - May 2023 *Remote*

- Engineered RESTful APIs ensuring data consistency and concurrency safety.
- Designed, architected, and implemented a custom authentication flow based on Auth0 and React overcoming embedded login limitations.

Full Stack Developer | Jan. 2021 - Jul. 2021 *Remote*

- Led frontend development of a digitalized gift card platform ensuring sub-3s rendering time for over 500 users.
- Designed and developed full-stack web applications with React, Node, and Serverless.

HONOURS & AWARDS

Best Final Year Research Project **Nov. 2023**

Obtained the award for the most outstanding work in the final year project of my graduating class (94%)

Dr. Gamini Wickremasinghe Gold Medal for Outstanding Achievement **Nov. 2023**

Obtained the award for achieving the highest average of my graduating class (89.3%)

BCS | National Best Quality Software Awards (NBQSA) **Oct. 2023**

Obtained a merit award for the category "tertiary student project technology" for introducing the LTS

PUBLICATIONS

IEEE | Annual Computing and Communication Workshop and Conference (CCWC) **Dec. 2023**

Proposed a novel neural network architecture, Liquid Time-Stochasticity (LTS), combining liquid neural networks with stochastic processes to improve forecasting in uncertain time-series environments.

ADDITIONAL WORK

Freelance **Nov. 2020 – May 2023**

- Developed an end-to-end diabetic retinopathy detection system using TensorFlow and CNNs, achieving 85+% sensitivity / specificity benchmarked against clinical screening thresholds.
- Architected a comprehensive multi-disease diagnostic platform for skin, lung, and breast cancer detection using deep learning models, enabling early-stage diagnosis and prognosis, implementing XAI visualizations for validation and trust.

EDUCATION

University of Westminster (IIT) **Nov. 2023**

BSc (Hons) Computer Science, First Class; Sergeant-At-Arms of IIT Toastmasters' *London*, United Kingdom

CERTIFICATIONS

Cloud: AWS Solutions Architect Associate

AI/ML: Machine Learning, Deep Learning, TensorFlow Developer, MLOps, AI for Medicine

SKILLS & INTERESTS

Languages: Python, TypeScript, JavaScript, Java

AI/ML: TensorFlow, Computer Vision, Forecasting, Gen AI

Backend: Node.js, Express.js, NestJS, Fast API

Cloud & DevOps: AWS, Docker & Kubernetes

Frontend: React, Next.js, Angular

Interests: AI/ML Research, MLOps, Gen AI, XAI, "breaking the AI black box", learning, teaching